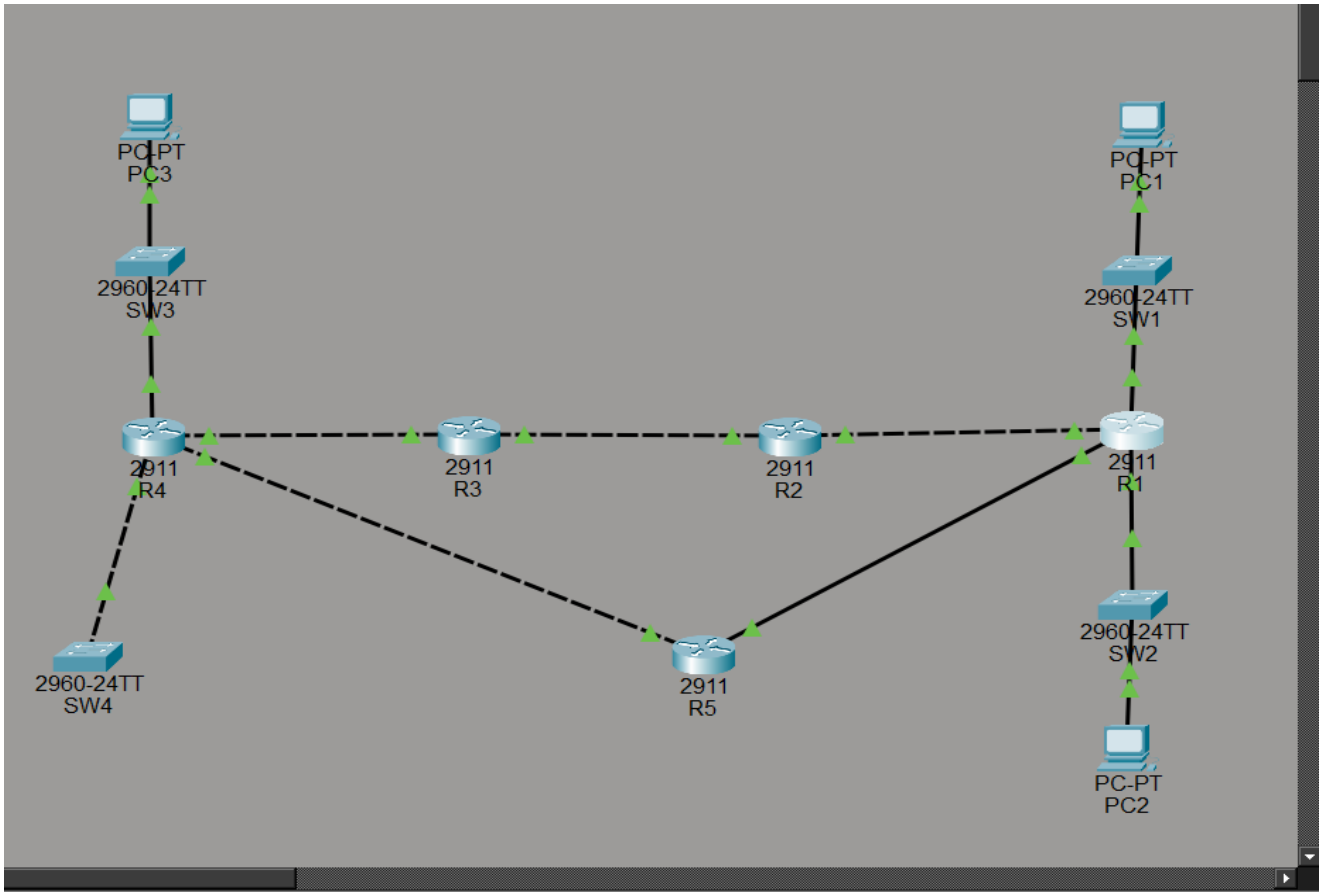
Week 7: Routing Fundamentals



Configure Hostnames

Routers

```
R1>enable
R1#show running-config | include hostname
hostname R1
R1#
```

```
R2>enable
R2#show running-config | include hostnames
R2#show running-config | include hostname
hostname R2
R2#
```

```
R3>enable
R3#show running-config | include hostname
hostname R3
R3#
```

```
hostname R4
R4#show running-config | include hostname
hostname R4
R4#
```

```
R5>
R5>enable
R5#show running-config | include hostname
hostname R5
R5#
```

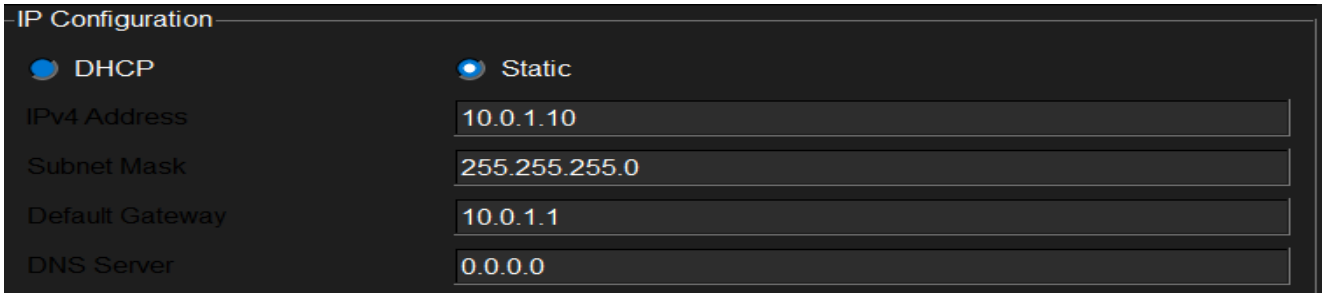
Configure IP Addresses on R1 ONLY

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#interface g0/2
R1(config-if)# ip address 10.0.0.1 255.255.255.0
% 10.0.0.0 overlaps with GigabitEthernet0/0
R1(config-if)# no shut
R1(config-if)# exit
R1(config)#
R1(config)#interface g0/0
R1(config-if)# ip address 10.0.1.1 255.255.255.0
% 10.0.1.0 overlaps with GigabitEthernet0/2
R1(config-if)# no shut
R1(config-if)# exit
R1(config)#
R1(config)#interface g0/1
R1(config-if)# ip address 10.0.3.1 255.255.255.0
R1(config-if)# no shut
R1(config-if)# exit
R1(config)#
R1(config)#interface vlan 2
R1(config-if)# ip address 10.0.2.1 255.255.255.0
R1(config-if)# no shut
R1(config-if)# exit
R1(config)#
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

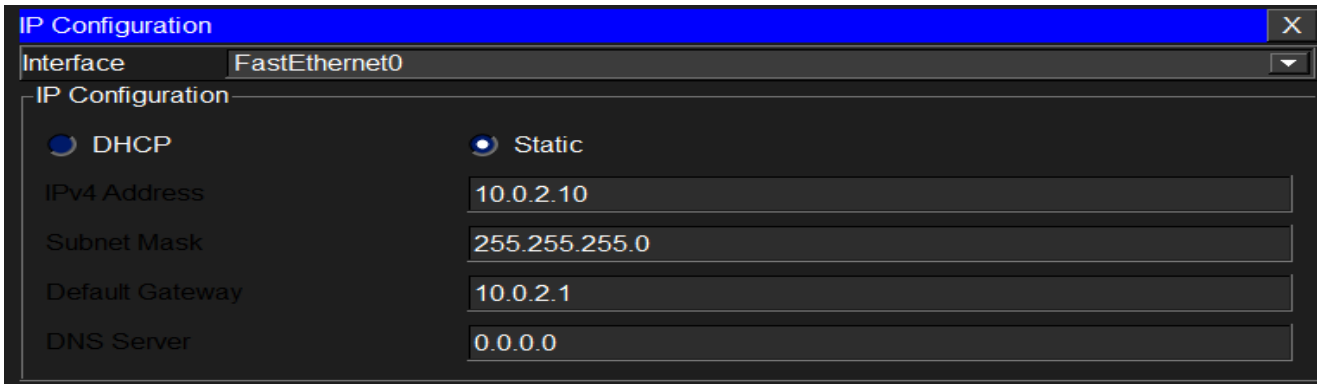
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on
FastEthernet0/3/1 (2), with Switch FastEthernet0/2 (1).

R1#
```

PC1



PC2



Verify Connected and Local Routes

On R1:

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
C       10.0.1.0/24 is directly connected, GigabitEthernet0/2
L       10.0.1.1/32 is directly connected, GigabitEthernet0/2
C       10.0.2.0/24 is directly connected, Vlan2
L       10.0.2.1/32 is directly connected, Vlan2
C       10.0.3.0/24 is directly connected, GigabitEthernet0/1
L       10.0.3.1/32 is directly connected, GigabitEthernet0/1

R1#
```

Ping PC2 From PC1

From PC1:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.2.10

Pinging 10.0.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```
C:\>tracert 10.0.2.10

Tracing route to 10.0.2.10 over a maximum of 30 hops:

  1    *        *        *          Request timed out.
  2    *        *        *          Request timed out.
  3    *        *        *          Request timed out.
  4    *        *        *          Request timed out.
  5    *        *        *          Request timed out.
  6    *        *        *          Request timed out.
  7    *        *        *          Request timed out.
  8    *        *        *          Request timed out.
  9    *        *        *          Request timed out.
  10    *        *        *         Request timed out.
  11    *        *        *         Request timed out.
  12    *        *        *         Request timed out.
  13    *        *        *         Request timed out.
  14    *        *        *         Request timed out.
  15    *        *        *         Request timed out.
  16    *        *        *         Request timed out.
  17    *        *        *         Request timed out.
  18    *        *        *         Request timed out.
  19    *        *        *         Request timed out.
  20    *        *        *         Request timed out.
  21    *        *        *         Request timed out.
  22    *        *        *         Request timed out.
  23    *        *        *         Request timed out.
  24    *        *        *         Request timed out.
  25    *        *        *         Request timed out.
  26    *        *        *         Request timed out.
  27    *        *        *         Request timed out.
  28    *        *        *         Request timed out.
  29    *        *        *         Request timed out.
  30    *        *        *         Request timed out.

Trace complete.
```

Ping PC3

From PC1:

```
C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

```
C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Request timed out.
```

Note: PC3 was already configured

Configure R2, R3 and R4

R2

```
R2>enable
R2#enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#hostname R2
R2(config)#
R2(config)#interface g0/0
R2(config-if)# ip address 10.0.0.2 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)#
R2(config)#interface g0/1
R2(config-if)# ip address 10.1.0.2 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)#
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#
```

R3

```
R3>enable
R3#enable
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#hostname R3
R3(config)#
R3(config)#interface g0/0
R3(config-if)# ip address 10.1.0.1 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)#
R3(config)#interface g0/1
R3(config-if)# ip address 10.1.1.2 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)#
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#
```

R4

```
Router>enable
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R4
R4(config)#
R4(config)#interface g0/0
R4(config-if)# ip address 10.1.1.1 255.255.255.0
R4(config-if)# no shutdown

R4(config-if)# exit
R4(config)#
R4(config)#interface g0/1
R4(config-if)# ip address 10.1.3.1 255.255.255.0
R4(config-if)# no shutdown

R4(config-if)# exit
R4(config)#
R4(config)#interface g0/2
R4(config-if)# ip address 10.1.2.1 255.255.255.0
R4(config-if)# no shutdown

R4(config-if)# exit
R4(config)#
R4(config)#end
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2,
changed state to up

R4#
%SYS-5-CONFIG_I: Configured from console by console
```

Verify PC3 Can Reach Its Gateway

From PC3:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.1.2.1

Pinging 10.1.2.1 with 32 bytes of data:

Reply from 10.1.2.1: bytes=32 time=55ms TTL=255
Reply from 10.1.2.1: bytes=32 time<1ms TTL=255
Reply from 10.1.2.1: bytes=32 time=4ms TTL=255
Reply from 10.1.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.1.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 55ms, Average = 14ms
```

Configure Static Routes

R1

```
R1>enable
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#ip route 10.1.0.0 255.255.255.0 10.0.0.2
R1(config)#ip route 10.1.1.0 255.255.255.0 10.0.0.2
R1(config)#ip route 10.1.2.0 255.255.255.0 10.0.0.2
R1(config)#ip route 10.1.3.0 255.255.255.0 10.0.0.2
R1(config)#
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

R2

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#
R2(config)#ip route 10.0.1.0 255.255.255.0 10.0.0.1
R2(config)#ip route 10.0.2.0 255.255.255.0 10.0.0.1
R2(config)#ip route 10.0.3.0 255.255.255.0 10.0.0.1
R2(config)#
R2(config)#ip route 10.1.1.0 255.255.255.0 10.1.0.1
R2(config)#ip route 10.1.2.0 255.255.255.0 10.1.0.1
R2(config)#ip route 10.1.3.0 255.255.255.0 10.1.0.1
R2(config)#
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
```

R3

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#
R3(config)#ip route 10.0.0.0 255.255.255.0 10.1.0.2
R3(config)#ip route 10.0.1.0 255.255.255.0 10.1.0.2
R3(config)#ip route 10.0.2.0 255.255.255.0 10.1.0.2
R3(config)#ip route 10.0.3.0 255.255.255.0 10.1.0.2
R3(config)#
R3(config)#ip route 10.1.2.0 255.255.255.0 10.1.1.1
R3(config)#ip route 10.1.3.0 255.255.255.0 10.1.1.1
R3(config)#
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console
```

R4

```
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#
R4(config)#ip route 10.0.0.0 255.255.255.0 10.1.1.2
R4(config)#ip route 10.0.1.0 255.255.255.0 10.1.1.2
R4(config)#ip route 10.0.2.0 255.255.255.0 10.1.1.2
R4(config)#ip route 10.0.3.0 255.255.255.0 10.1.1.2
R4(config)#
R4(config)#ip route 10.1.0.0 255.255.255.0 10.1.1.2
R4(config)#
R4(config)#end
R4#
%SYS-5-CONFIG_I: Configured from console by console
```

Verify Connectivity

From PC1:

```
C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```
C:\>tracert 10.1.2.10

Tracing route to 10.1.2.10 over a maximum of 30 hops:

  1    *        *        *          Request timed out.
  2    *        *        *          Request timed out.
  3    *        *        *          Request timed out.
  4    *        *        *          Request timed out.
  5    *        *        *          Request timed out.
  6    *        *        *          Request timed out.
  7    *        *        *          Request timed out.
  8    *        *        *          Request timed out.
  9    *        *        *          Request timed out.
  10    *        *        *          Request timed out.
  11    *        *        *          Request timed out.
  12    *        *        *          Request timed out.
  13    *        *        *          Request timed out.
  14    *        *        *          Request timed out.
  15    *        *        *          Request timed out.
  16    *        *        *          Request timed out.
  17    *        *        *          Request timed out.
  18    *        *        *          Request timed out.
  19    *        *        *          Request timed out.
  20    *        *        *          Request timed out.
  21    *        *        *          Request timed out.
  22    *        *        *          Request timed out.
  23    *        *        *          Request timed out.
  24    *        *        *          Request timed out.
  25    *        *        *          Request timed out.
  26    *        *        *          Request timed out.
  27    *        *        *          Request timed out.
  28    *        *        *          Request timed out.
  29    *        *        *          Request timed out.
  30    *        *        *          Request timed out.

Trace complete.
```

Remove All Static Routes from R1

```
R1>enable
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#no ip route 10.1.0.0 255.255.255.0 10.0.0.2
R1(config)#no ip route 10.1.1.0 255.255.255.0 10.0.0.2
R1(config)#no ip route 10.1.2.0 255.255.255.0 10.0.0.2
R1(config)#no ip route 10.1.3.0 255.255.255.0 10.0.0.2
R1(config)#
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

Test

From PC1:

```
C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Request timed out.
```

Add One Summary Route

```
R1#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on
FastEthernet0/3/1 (2), with Switch FastEthernet0/2 (1).
conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#ip route 10.1.0.0 255.255.252.0 10.0.0.2
R1(config)#
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#
```

Verify Summary Route

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 7 subnets, 3 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
C       10.0.1.0/24 is directly connected, GigabitEthernet0/2
L       10.0.1.1/32 is directly connected, GigabitEthernet0/2
C       10.0.2.0/24 is directly connected, Vlan2
L       10.0.2.1/32 is directly connected, Vlan2
S       10.1.0.0/22 [1/0] via 10.0.0.2
```

Test Connectivity

From PC1

```
C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Configure R5

```
Router#enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface g0/0
Router(config-if)# ip address 10.1.3.2 255.255.255.0
Router(config-if)# no shutdown

Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)# ip address 10.0.3.2 255.255.255.0
Router(config-if)# no shutdown

Router(config-if)#
Router(config-if)#end
Router#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Check Connectivity

From PC1:

```
C:\>ping 10.1.3.2

Pinging 10.1.3.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.1.3.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Configure Longest Prefix Match

On R1:

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#ip route 10.1.3.0 255.255.255.0 10.0.3.2
R1(config)#
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

Verify New Path

From PC1:

```
C:\>tracert 10.1.3.2

Tracing route to 10.1.3.2 over a maximum of 30 hops:

  1    *          *          *          Request timed out.
  2    *          *          *          Request timed out.
  3    *          *          *          Request timed out.
  4    *          *          *          Request timed out.
  5    *          *          *          Request timed out.
  6    *          *          *          Request timed out.
  7    *          *          *          Request timed out.
  8    *          *          *          Request timed out.
  9    *          *          *          Request timed out.
  10    *          *          *          Request timed out.
  11    *          *          *          Request timed out.
  12    *          *          *          Request timed out.
  13    *          *          *          Request timed out.
  14    *          *          *          Request timed out.
  15    *          *          *          Request timed out.
  16    *          *          *          Request timed out.
  17    *          *          *          Request timed out.
  18    *          *          *          Request timed out.
  19    *          *          *          Request timed out.
  20    *          *          *          Request timed out.
  21    *          *          *          Request timed out.
  22    *          *          *          Request timed out.
  23    *          *          *          Request timed out.
  24    *          *          *          Request timed out.
  25    *          *          *          Request timed out.
  26    *          *          *          Request timed out.
  27    *          *          *          Request timed out.
  28    *          *          *          Request timed out.
  29    *          *          *          Request timed out.
  30    *          *          *          Request timed out.

Trace complete.
```

Configure R4 Internet Interface

```
R4>en
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#
R4(config)#interface vlan 10
R4(config-if)# ip address 203.0.113.1 255.255.255.0
R4(config-if)# no shutdown
R4(config-if)# exit
R4(config)#
R4(config)#interface fa0/3/0
R4(config-if)# switchport access vlan 10
R4(config-if)# no shutdown
R4(config-if)#
R4(config-if)#end
R4#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to
up

%SYS-5-CONFIG_I: Configured from console by console
```

Configure a Default Route

R1

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.2
R1(config)#ip route 0.0.0.0 0.0.0.0 10.0.3.2
R1(config)#
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

R2

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.2
R1(config)#ip route 0.0.0.0 0.0.0.0 10.0.3.2
R1(config)#
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

R3

```
R3>EN
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ip route 0.0.0.0 0.0.0.0 10.1.1.1
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console
```

R5

```
Router>EN
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 10.1.3.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Verify Load Balancing

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.0.2 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 10 subnets, 3 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
C       10.0.1.0/24 is directly connected, GigabitEthernet0/2
L       10.0.1.1/32 is directly connected, GigabitEthernet0/2
C       10.0.2.0/24 is directly connected, Vlan2
L       10.0.2.1/32 is directly connected, Vlan2
C       10.0.3.0/24 is directly connected, GigabitEthernet0/1
L       10.0.3.1/32 is directly connected, GigabitEthernet0/1
S       10.1.0.0/22 [1/0] via 10.0.0.2
S       10.1.3.0/24 [1/0] via 10.0.3.2
S*   0.0.0.0/0 [1/0] via 10.0.0.2
                 [1/0] via 10.0.3.2

R1#
R1#
```

Reflection Questions (Answers)

20. Do you see routes for every network R1 is directly connected to? Explain why or why not.

Yes. R1 automatically creates **connected (C)** routes for each active interface network and **local (L)** routes for each interface IP address. Connected routes allow the router to reach directly attached networks, while local routes identify the router's own interface addresses.

21. Why does PC1 lose reachability to PC3 once the static routes are removed from R1, even though PC1 and PC3 haven't changed?

PC1 still sends traffic to its default gateway (R1), but after the static routes are removed, R1 no longer knows how to reach the remote network **10.1.2.0/24** where PC3 resides. Since R1 has no matching route, it drops the packets, causing connectivity to fail.

22. Why is a single summary route able to replace multiple /24 static routes on R1?

The networks:

Plain Text

10.1.0.0/24

10.1.1.0/24

10.1.2.0/24

10.1.3.0/24

are contiguous and can be summarized as:

Plain Text

10.1.0.0/22

Route represents all four networks with one routing-table entry, reducing routing-table size and simplifying administration while maintaining connectivity.